MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC ASSESSMENT TEST – 11** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 4: Quadratic Equations** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, zero skipped steps, and final boxed answers with proper units.
2. In alternative (**OR**) sub-questions, complete step-by-step solutions and equal step marking apply to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Roots rejected by physical domain/restrictions are explicitly stated.

Official Evaluation Key | Maths Manthan Meerut

Section A: MCQ & Assertion-Reason Questions

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Q1. If the quadratic equation $x^2 + k(4x + k - 1) + 2 = 0$ has equal roots, then $k =$: [1]

(A) $-\frac{2}{3},\ 1$ **(B)** $\frac{2}{3},\ -1$

(C) $\frac{3}{2},\ \frac{1}{3}$ **(D)** $-\frac{1}{3},\ \frac{2}{3}$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $\frac{2}{3},\ -1$

Writing $x^2 + 4kx + (k^2 - k + 2) = 0$ in standard form, we have $a = 1$, $b = 4k$, and $c = k^2 - k + 2$.

Formula Used: Equal real roots $\iff D = b^2 - 4ac = 0$

$$(4k)^2 - 4(1)(k^2 - k + 2) = 0$$

$$16k^2 - 4k^2 + 4k - 8 = 0 \implies 12k^2 + 4k - 8 = 0 \implies 3k^2 + k - 2 = 0$$

Factorising $(3k^2 + 3k - 2k - 2 = 0)$:

$$3k(k + 1) - 2(k + 1) = 0 \implies (3k - 2)(k + 1) = 0 \implies \boxed{k = \frac{2}{3},\ -1}$$

Hence, option **(B)** is correct. **[1 mark]**

Q2. If the roots of the equation $(a^2+b^2)x^2-2b(a+c)x+(b^2+c^2)=0$ are equal, then: [1]

(A) $2b=a+c$ **(B)** $b=\dfrac{2ac}{a+c}$

(C) $b^2=ac$ **(D)** $b=ac$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(C)** $b^2=ac$

Comparing with $Ax^2+Bx+C=0$, $A=a^2+b^2$, $B=-2b(a+c)$, and $C=b^2+c^2$.

Formula Used: Equal real roots $\iff D=B^2-4AC=0$

$$4b^2(a+c)^2-4(a^2+b^2)(b^2+c^2)=0$$
$$b^2(a^2+2ac+c^2)-(a^2b^2+a^2c^2+b^4+b^2c^2)=0$$
$$a^2b^2+2ab^2c+b^2c^2-a^2b^2-a^2c^2-b^4-b^2c^2=0$$
$$-(b^4-2ab^2c+a^2c^2)=0 \implies (b^2-ac)^2=0 \implies \boxed{b^2=ac}$$

Hence, option **(C)** is correct. **[1 mark]**

Q3. If the equation $ax^2+2x+a=0$ has two equal roots, then $a=$: [1]

(A) ±1 **(B)** ±2

(C) 0 **(D)** $\pm\dfrac{1}{2}$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(A)** ±1

Since $ax^2+2x+a=0$ is quadratic, $a\neq0$, and for equal roots $D=0$:

Formula Used: $D=b^2-4ac=0$

$$(2)^2-4(a)(a)=0 \implies 4-4a^2=0 \implies a^2=1 \implies \boxed{a=\pm1}$$

Hence, option **(A)** is correct. **[1 mark]**

Q4. If the quadratic equation $x^2-bx+1=0$ does not possess real roots, then: [1]

(A) $-3<b<3$ **(B)** $-2<b<2$

(C) $b>2$ **(D)** $b<-2$

▶ STEP-BY-STEP SOLUTION: [1 Mark]

Correct Option: **(B)** $-2<b<2$

Comparing $x^2-bx+1=0$ with $Ax^2+Bx+C=0$, we have $A=1$, $B=-b$, and $C=1$.

Formula Used: No real roots $\iff D=B^2-4AC<0$

$$(-b)^2-4(1)(1)<0 \implies b^2-4<0 \implies b^2<4 \implies \boxed{-2<b<2}$$

Hence, option **(B)** is correct. **[1 mark]**

Q5. If a and b are the roots of the quadratic equation $x^2 + ax + b = 0$, then $a + b =$: [1]

(A) -1 **(B)** 1
(C) 2 **(D)** -2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A) -1

Here a and b are the roots of $x^2 + ax + b = 0$, assuming distinct/non-trivial quadratic roots (note if $b = 0$ then $a = 0 \implies a + b = 0$, which is not an option; for $b \neq 0$):

Formula Used: Sum: $a + b = -a$, Product: $ab = b$

From the product $ab = b \implies b(a - 1) = 0 \implies a = 1$ (since $a = b = 0$ gives 0, not in options).
Substituting $a = 1$ into $a + b = -a$:

$$1 + b = -1 \implies b = -2 \quad \text{Therefore, } a + b = 1 + (-2) = \boxed{-1}$$

(Check: roots of $x^2 + x - 2 = 0$ are $1, -2$.) Hence, option **(A)** is correct. **[1 mark]**

Q6. Assertion-Reason: [1]

Directions: *(A) Both A & R true, R explains A; (B) Both true, R does NOT explain A; (C) A true, R false; (D) A false, R true.*

Assertion (A): A quadratic equation cannot possess more than two real roots.
Reason (R): Fundamental Theorem of Algebra guarantees that a polynomial of degree n has at most n roots.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) Both Assertion (A) and Reason (R) are true, and Reason (R) is the correct explanation of Assertion (A).**

A quadratic equation $ax^2 + bx + c = 0$ $(a \neq 0)$ is a polynomial equation of degree $n = 2$.

Formula Used: Degree $n = 2 \implies$ at most 2 roots

By Reason (R), a degree-2 polynomial can have at most 2 complex (and hence at most 2 real) roots (0, 1 repeated, or 2 distinct real roots).
Thus Assertion (A) is **true**, Reason (R) is **true**, and Reason (R) correctly explains Assertion (A). **[1 mark]**

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Q7. Solve the following quadratic equation for x by factorization: $2x^2 + ax - a^2 = 0$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given equation: $2x^2 + ax - a^2 = 0$. Here coefficient of x^2 is 2, $b = a$, and $c = -a^2$, so $ac = -2a^2$.

Formula Used: Split $+ax = +2ax - ax$ since $(2a)(-a) = -2a^2$ and $2a - a = a$

$$\begin{aligned} 2x^2 + 2ax - ax - a^2 &= 0 \\ 2x(x + a) - a(x + a) &= 0 \quad \textbf{[1 mark]} \\ (x + a)(2x - a) &= 0 \end{aligned}$$

Equating each factor to zero:

$$x + a = 0 \implies x = -a \qquad \text{or} \qquad 2x - a = 0 \implies x = \frac{a}{2}$$

Hence, the roots are $\boxed{x = \frac{a}{2} \text{ and } x = -a}$. **[1 mark]**

Q8. Find the roots of the following quadratic equation: $x^2 - 2ax + (a^2 - b^2) = 0$. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Rewriting $x^2 - 2ax + (a^2 - b^2) = 0$ either by completing the square or factorising $(a - b)(a + b)$:

Formula Used: $A^2 - B^2 = (A - B)(A + B)$

Grouping the perfect square $x^2 - 2ax + a^2 = (x - a)^2$:

$$\begin{aligned} (x^2 - 2ax + a^2) - b^2 &= 0 \\ (x - a)^2 - b^2 &= 0 \quad \textbf{[1 mark]} \\ [(x - a) - b]\,[(x - a) + b] &= 0 \end{aligned}$$

Equating each factor to zero:

$$x - (a + b) = 0 \implies x = a + b \qquad \text{or} \qquad x - (a - b) = 0 \implies x = a - b$$

Hence, the roots are $\boxed{x = a + b \text{ and } x = a - b}$. **[1 mark]**

Q9. The sum of two numbers is 15. If the sum of their reciprocals is $\frac{3}{10}$, find the two numbers. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let one number be x. Since their sum is 15, the other number is $15 - x$ ($x \neq 0, 15$).

Formula Used: $\frac{1}{x} + \frac{1}{15-x} = \frac{3}{10}$

Simplifying the LHS over common denominator $x(15-x)$:

$$\frac{(15-x)+x}{x(15-x)} = \frac{3}{10}$$

$$\frac{15}{15x-x^2} = \frac{3}{10} \implies \frac{5}{15x-x^2} = \frac{1}{10} \quad \text{(dividing by 3)} \quad \textbf{[1 mark]}$$

Cross-multiplying:

$$15x - x^2 = 50 \implies x^2 - 15x + 50 = 0 \quad \textbf{[1 mark]}$$

Factorising ($-10 \times (-5) = 50$ and $-10 - 5 = -15$):

$$x^2 - 10x - 5x + 50 = 0 \implies (x-10)(x-5) = 0 \implies x = 10 \text{ or } x = 5$$

If $x = 10$, the other number is $15 - 10 = 5$; if $x = 5$, the other is $15 - 5 = 10$. (Check: $\frac{1}{5} + \frac{1}{10} = \frac{3}{10}$.)
Hence, the two numbers are $\boxed{5 \text{ and } 10}$. **[1 mark]**

Q10. A two-digit number is four times the sum of its digits and twice the product of its digits. Find the number. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Let the tens digit be x ($1 \leq x \leq 9$) and the units digit be y ($0 \leq y \leq 9$). Then the two-digit number $= 10x + y$.

Formula Used: $10x + y = 4(x + y)$ and $10x + y = 2xy$

From the first condition:

$$10x + y = 4x + 4y \implies 6x = 3y \implies y = 2x \quad \textbf{[1 mark]}$$

Substituting $y = 2x$ into the second condition $10x + y = 2xy$:

$$10x + 2x = 2x(2x)$$

$$12x = 4x^2 \implies 4x^2 - 12x = 0 \implies 4x(x-3) = 0 \quad \textbf{[1 mark]}$$

Since tens digit of a two-digit number cannot be zero ($x \neq 0$), we reject $x = 0$ and take $x = 3$.
Then units digit $y = 2(3) = 6$, giving the number $10(3) + 6 = 36$. (Check: $4(3+6) = 36$ and $2(3)(6) = 36$.)
Hence, the required two-digit number is $\boxed{36}$. **[1 mark]**

Q11. Two pipes running together can fill a cistern in $3\frac{1}{13}$ minutes. If one pipe takes 3 minutes more than the other to fill it, find the time in which each pipe would fill the cistern separately. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let the faster pipe take x minutes ($x > 0$) to fill the cistern alone.

Then the slower pipe takes $(x+3)$ minutes alone. Together they fill the cistern in $3\frac{1}{13} = \frac{40}{13}$ minutes. **[1 mark]**

Formula Used: Work in 1 min (Pipe 1) + Work in 1 min (Pipe 2) $= \frac{1}{\text{Combined Time}}$

Setting up the rate equation per minute:

$$\frac{1}{x} + \frac{1}{x+3} = \frac{13}{40}$$
$$\frac{(x+3)+x}{x(x+3)} = \frac{13}{40} \implies \frac{2x+3}{x^2+3x} = \frac{13}{40} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$

Cross-multiplying:

$$40(2x+3) = 13(x^2+3x) \implies 80x + 120 = 13x^2 + 39x \implies 13x^2 - 41x - 120 = 0 \quad \textbf{[1 mark]}$$

Factorising ($13 \times (-120) = -1560$; $-65 \times 24 = -1560$ and $-65 + 24 = -41$):

$$13x^2 - 65x + 24x - 120 = 0 \implies 13x(x-5) + 24(x-5) = 0 \implies (x-5)(13x+24) = 0 \quad \textbf{[1 mark]}$$

Thus $x = 5$ or $x = -\frac{24}{13}$. Since time cannot be negative ($x > 0$), we reject $x = -\frac{24}{13}$.

So faster pipe takes 5 minutes and slower pipe takes $5 + 3 = 8$ minutes. (Check: $\frac{1}{5} + \frac{1}{8} = \frac{13}{40}$.)

Hence, $\boxed{\text{Faster pipe} = 5 \text{ minutes, Slower pipe} = 8 \text{ minutes}}$. **[$\frac{1}{2}$ mark]**

Q12. At t minutes past 2 pm, the time needed by the minute hand of a clock to show 3 pm was found to be 3 minutes less than $\frac{t^2}{4}$ minutes. Find the value of t. **[5]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Since 3 pm is 60 minutes past 2 pm, at t minutes past 2 pm ($0 \le t < 60$), the actual time remaining until 3 pm is:

$$\text{Time needed} = (60 - t) \text{ minutes} \quad \textbf{[1 mark]}$$

Formula Used: Time remaining to 3 pm $= 60 - t = \frac{t^2}{4} - 3$

Equating both expressions for the remaining time:

$$60 - t = \frac{t^2}{4} - 3$$
$$240 - 4t = t^2 - 12 \quad \text{(multiplying throughout by 4)} \quad \textbf{[1}\tfrac{1}{2}\textbf{ marks]}$$
$$t^2 + 4t - 252 = 0 \quad \textbf{[1 mark]}$$

Factorising ($18 \times (-14) = -252$ and $18 - 14 = 4$):

$$t^2 + 18t - 14t - 252 = 0 \implies t(t+18) - 14(t+18) = 0 \implies (t+18)(t-14) = 0 \quad \textbf{[1 mark]}$$

This gives $t = -18$ or $t = 14$. Since t represents minutes past 2 pm ($0 \le t < 60$), we reject $t = -18$.

(Check: At $t = 14$ min, time to 3 pm $= 60 - 14 = 46$ min, and $\frac{14^2}{4} - 3 = 49 - 3 = 46$ min.)

Hence, the required value is $\boxed{t = 14}$. **[$\frac{1}{2}$ mark]**

Q13. Case Study: Railway Logistics & Freight Corridor Speed Optimization

A dedicated freight corridor connects an inland manufacturing cluster to an international maritime port located 360 km away. A heavy container cargo train travels at a scheduled average velocity of v km/h. Logistics engineers compute that increasing the average velocity by 5 km/h would save exactly 1 hour on the 360 km transit. [4]

(i) Write algebraic expressions for transit time at velocity v and velocity $(v+5)$ over the 360 km track. **[1]**

(ii) Formulate the quadratic equation in v in standard form representing the transit time reduction. **[1]**

(iii) Determine the scheduled average velocity of the container train (v). **OR** Calculate the total operating time saved per month if the freight train operates 60 scheduled runs at the optimized higher velocity. **[2]**

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

We know the relation between distance, velocity, and transit time:

Formula Used: $\text{Time} = \dfrac{\text{Distance}}{\text{Velocity}}$

(i) Transit Time Expressions: For a 360 km haul at scheduled velocity v km/h ($v > 0$) and optimized velocity $(v+5)$ km/h:

$$\boxed{t_1 = \frac{360}{v} \text{ hours}} \quad \text{and} \quad \boxed{t_2 = \frac{360}{v+5} \text{ hours}} \quad \textbf{[1 mark]}$$

(ii) Quadratic Equation in Standard Form: Increasing velocity by 5 km/h saves 1 hour ($t_1 - t_2 = 1$):

$$\frac{360}{v} - \frac{360}{v+5} = 1$$

$$360\left[\frac{(v+5)-v}{v(v+5)}\right] = 1 \implies \frac{1800}{v^2+5v} = 1 \implies \boxed{v^2+5v-1800=0} \quad \textbf{[1 mark]}$$

(iii) First Part (Scheduled Velocity v): Splitting the middle term ($45 \times (-40) = -1800$ and $45 - 40 = 5$):

$$v^2 + 45v - 40v - 1800 = 0 \implies v(v+45) - 40(v+45) = 0$$
$$\implies (v+45)(v-40) = 0 \implies v = -45 \text{ or } v = 40$$

Since train velocity cannot be negative ($v > 0$), we reject $v = -45$.
Hence, the scheduled average velocity of the container train is $\boxed{40 \text{ km/h}}$. **[2 marks]**

OR (Alternative Part — Monthly Operating Time Saved): Each single optimized run saves 1 hour ($\frac{360}{40} - \frac{360}{45} = 9 - 8 = 1$ h).
For 60 scheduled runs per month at the optimized higher velocity (45 km/h):

$$\text{Total time saved per month} = 60 \times 1 \text{ hour} = \boxed{60 \text{ hours}} \quad \textbf{[2 marks]}$$

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